

1. k-means. 2. a. gradient descent b. gradient ascent 3. polynomial $k(x_i, x_j) = (1 + x_i x_j)^P$
4. $P(B|C)$ 5. $P(y|x_1, x_2, \dots, x_n) = \frac{P(y) \cdot P(x_1|y) \cdot P(x_2|y) \cdot \dots \cdot P(x_n|y)}{P(x_1, x_2, \dots, x_n)} = P(y) \cdot \prod_{i=1}^n P(x_i|y)$
6. drop out. 7. choose the cluster centers with the greatest distance.
8. 9. mean squared error (MSE) 10. silhouette score
11. $\frac{FP}{FP+TN}$, $\frac{2 \cdot P \cdot R}{P+R}$ 12. a. underfitting b. overfitting
13. a. F b. F c. F d. T e. F f. F g. T h. T i. F j. T

Long form

1. $W_1 = [0, 0]$. $W_1 \cdot X_1 = [0, 0] \cdot [2, 1] = 0 \rightarrow \text{predict } +$
 $W_1 \cdot X_2 = [0, 0] \cdot [2, 3] = 0 \rightarrow +$
 $W_1 \cdot X_3 = 0 \rightarrow \text{incorrect}, -$
- $W_2 = W_1 - X_3 = [0, 0] - [4, 0] = [-4, 0]$. $W_2 \cdot X_4 = [-4, 0] \cdot [1, -2] = 4 \rightarrow + \text{ incorrect}$
 $W_3 = W_2 - X_4 = [-4, 0] - [1, -2] = [-3, 2]$. $W_3 \cdot X_5 = [-3, 2] \cdot [1, 2] = 7 \rightarrow + \text{ correct}$
 $W_3 \cdot X_6 = [-3, 2] \cdot [-3, -2] = -13 \rightarrow - \text{ correct}$
 $W_3 \cdot X_1 = [-3, 2] \cdot [2, 1] = -8 \rightarrow +$ $W_3 \cdot X_2 = [-3, 2] \cdot [2, 3] = 0 \rightarrow +$
 $W_3 \cdot X_3 = [-3, 2] \cdot [4, 0] = -12 \rightarrow -$, $W_3 \cdot X_4 = [-3, 2] \cdot [1, -2] = -1 \rightarrow -$
- Linear separator: $3x_1 + 2x_2 = 0$

2. $P(\text{Drug A}) = \frac{5}{14} = 0.357$ $P(\text{Drug B}) = \frac{9}{14} = 0.643$
 $P(\text{Young} | \text{Drug A}) = \frac{3+1}{5+3} = \frac{4}{8} = 0.5$ $P(\text{Young} | \text{Drug B}) = \frac{2+1}{9+3} = \frac{3}{12} = 0.25$
 $P(\text{Male} | \text{Drug A}) = \frac{1+1}{5+3} = \frac{2}{8} = 0.25$ $P(\text{Male} | \text{Drug B}) = \frac{6+1}{9+3} = \frac{7}{12} = 0.583$
 $P(\text{High} | \text{Drug A}) = \frac{2+1}{5+3} = \frac{3}{8} = 0.375$ $P(\text{High} | \text{Drug B}) = \frac{2+1}{9+3} = \frac{3}{12} = 0.25$
 $P(\text{Unsafe} | \text{Drug A}) = \frac{5+3}{5+3} = 1$ $P(\text{Unsafe} | \text{Drug B}) = \frac{3+1}{9+3} = \frac{4}{12} = 0.333$

$$P(\text{Drug A} | \text{input}) = 0.357 \times 0.5 \times 0.25 \times 0.375 \times 1 = 0.0169$$

$$P(\text{Drug B} | \text{input}) = 0.643 \times 0.25 \times 0.583 \times 0.25 \times 0.333 = 0.0093$$

$$\text{argmax} \{ \text{Drug A}, \text{Drug B} \} = \text{Drug A}, P(\text{Drug A} | \text{instance}) = \frac{0.0169}{0.0169 + 0.0093} = 0.64$$

3. $P(+)=\frac{2}{6}=0.33$, $P(-)=\frac{0.087}{0.087}, P(N)=0.33$, use Laplace smoothing $\alpha=1$
 $P(\text{winter} | +) = \frac{1+1}{9+14} = \frac{2}{23} = 0.087$, $P(\text{winter} | N) = \frac{1+1}{9+14} = \frac{2}{23} = 0.087$, $P(\text{winter} | -) = \frac{0+1}{9+14} = \frac{1}{23} = 0.043$
 $P(\text{is} | +) = \frac{2+1}{9+14} = \frac{3}{23} = 0.13$, $P(\text{is} | N) = \frac{2+1}{9+14} = \frac{3}{23} = 0.13$, $P(\text{is} | -) = \frac{2+1}{9+14} = \frac{3}{23} = 0.13$
 $P(\text{not} | +) = \frac{0+1}{9+14} = \frac{1}{23} = 0.043$, $P(\text{not} | N) = \frac{0+1}{9+14} = \frac{1}{23} = 0.043$, $P(\text{not} | -) = \frac{1+1}{9+14} = \frac{2}{23} = 0.087$
 $P(\text{pleasant} | +) = \frac{1+1}{9+14} = \frac{2}{23} = 0.087$, $P(\text{pleasant} | N) = \frac{0+1}{9+14} = \frac{1}{23} = 0.043$, $P(\text{pleasant} | -) = \frac{0+1}{9+14} = \frac{1}{23} = 0.043$
 $P(+ | \text{input}) = 0.33 \times 0.087 \times 0.13 \times 0.043 \times 0.087 = 1.39625 \times 10^{-5}$
 $P(N | \text{input}) = 0.33 \times 0.087 \times 0.13 \times 0.043 \times 0.043 = 6.90102 \times 10^{-6}$
 $P(- | \text{input}) = 0.33 \times 0.043 \times 0.13 \times 0.087 \times 0.043 = 6.90102 \times 10^{-6}$
The output class is estimated to be positive

$$4. \quad [w_1, w_2, w_3] = [-0.3, 0.7, 0.5] \\ \theta_0 = [-0.3, 0.7, 0.5], \quad \theta_1 = \theta_0 - \left(\frac{y}{n}\right) \frac{\partial L}{\partial \theta} = \theta_0 - \left(\frac{y}{n}\right) \sum_{i=1,2,3} x_{ij} \cdot [\theta^T x - y_i]$$

$$\theta^T x = [w_1, w_2, w_3] \cdot [x_1, x_2, x_3]$$

$$\theta^T x_1 = [-0.3, 0.7, 0.5] \cdot [-1, -3, 2] = -0.8, \quad \theta^T x_1 - y_1 = 0.5$$

$$\theta^T x_2 = [-0.3, 0.7, 0.5] \cdot [0, 2, -1] = 0.9, \quad \theta^T x_2 - y_2 = -0.8$$

$$\theta^T x_3 = [-0.3, 0.7, 0.5] \cdot [3, -1, 1] = -1.1, \quad \theta^T x_3 - y_3 = -3.6$$

$$\frac{\partial L}{\partial w_1} = \sum_{i=1}^3 x_{i1} [\theta^T x_i - y_i] = 0.5 \times (-1) + (-0.8) \times 0 + (-3.6) \times 3 = -11.3 \Rightarrow \frac{y}{n} \cdot \frac{\partial L}{\partial w_1} = -3.77$$

$$\frac{\partial L}{\partial w_2} = \sum_{i=1}^3 x_{i2} [\theta^T x_i - y_i] = 0.5 \times (-3) + (-0.8) \times 2 + (-3.6) \times (-1) = 0.5 \Rightarrow \frac{y}{n} \cdot \frac{\partial L}{\partial w_2} = 0.167$$

$$\frac{\partial L}{\partial w_3} = \sum_{i=1}^3 x_{i3} [\theta^T x_i - y_i] = 0.5 \times (2) + (-0.8) \times (-1) + (-3.6) \times 1 = -1.8 \Rightarrow \frac{y}{n} \cdot \frac{\partial L}{\partial w_3} = -0.6$$

$$w' = [w_1, w_2, w_3] - [-3.77, 0.167, -0.6] = [3.467, 0.533, 1.1]$$

Predict output ~~for~~ with new weights.

$$\hat{y}_1 = w'^T x_1 = [3.467, 0.533, 1.1] \cdot [-1, -3, 2] = -2.866$$

$$\hat{y}_2 = w'^T x_2 = [3.467, 0.533, 1.1] \cdot [0, 2, -1] = -0.034$$

$$\hat{y}_3 = w'^T x_3 = [3.467, 0.533, 1.1] \cdot [3, -1, 1] = 10.968$$

$$RMSE = \sqrt{\frac{1}{3} \sum_{i=1}^3 (\hat{y}_i - y_i)^2} = \sqrt{\frac{1}{3} [(-2.866 + 1.3)^2 + (-0.034 - 1.7)^2 + (10.968 - 2.5)^2]} = 5.07$$